VUG(I)-BCA (F-1001)

2023-26 (Voc)

Full Marks : 70

Time : 3 hours

Answer from **both** the Sections as directed.

The figures in the right-hand margin indicate marks.

Candidates are required to give their answers in their own words as far as practicable.

SECTION–A

Answer any *four* questions from the following: 10×4

1. Explain the principles of effective communication. Discuss how barriers to communication can be overcome. 4 + 6

2. What do you mean by feedback ? Discuss the importance of feedback in listening. 2 + 8

3. (*i*) Correct following as directed: 5

   (*a*) He hanged the coat on the peg. (Correct the wrong word)

(*b*) Zubeen Garg is an imminent singer. (Correct the wrong word)

(*c*) Earthly pleasures are momentary / momentous. (Choose the Correct option)

(*d*) The Sun is stationary / stationery. (Choose the Correct option)

(*e*) The minister will visit shortly / recently. (Choose the Correct option)

(*ii*) Do as directed: 5

(*a*) Study of the human mind. (Express in one word)

(*b*) Incapable of being seen through. (Express in one word)

(*c*) Give the antonym of 'Prosperity'.

(*d*) Give the synonym of 'Bogus'.

(*e*) Promises should be kept. (Change the voice)

(Continued)

4. Fill in the blanks with the correct use of tenses. 10

(*a*) My mother _____ up early in the morning.

(*i*) get

(*ii*) gets

(*iii*) will be

(*iv*) shall be

(*b*) The children _____ in the field now.

(*i*) has played

(*ii*) are playing

(*iii*) plays

(*iv*) will had played

(*c*) I _____ her for several years.

(*i*) has known

(*ii*) have known

(*iii*) knows

(*iv*) knew

(*d*) It _____ raining since morning.

(*i*) have been

(*ii*) has been

(*iii*) is

(*iv*) was

(*e*) He _____ his house seven days ago.

(*i*) left

(*ii*) leave

(*iii*) leaves

(*iv*) is leaving

(*f*) I _____ a letter when he came to my house.

(*i*) am writing

(*ii*) was writing

(*iii*) will write

(*iv*) wrote

(*g*) The train _____ before he reached the station.

(*i*) has left

(*ii*) was left

(*iii*) had left

(*iv*) is left

(*h*) It _____ raining since morning when you rang me up.

(*i*) had been

(*ii*) has been

(*iii*) have been

(*iv*) was

(*i*) If he works hard he _____ pass.

(*i*) will

(*ii*) shall

(*iii*) will be

(*iv*) shall be

(*j*) The cook _____ cooking food at this time tomorrow.

(*i*) shall be

(*ii*) will be

(*iii*) shall

(*iv*) will

5. What is an office order ? Mention some of the occasions in which office orders are issued. Draft an office order asking employees to avoid smoking during office hours.

2 + 2 + 6

6. You are a purchase manager of Sipla Marketing Agencies. Write a letter to Thompson Matrix Ltd., Bangalore asking for a copy of their latest catalogue of home appliances. Ask for best trade terms assuring long term business relationship.

7. Prepare a resume for a Computer Sc. Graduate, with some work experience, aspiring for the post of a Programmer in a reputed company.

8. What is importance of Group Discussion? Discuss different types of GD with examples.

SECTION–B

Answer *all* questions from the following; 3×10

9. Briefly Explain objectives of business communication.

10. Differentiate between Listening and Hearing.

11. Define minutes and resolution.

12. Write short note on two-way model of communication.

13. Explain the Postures and Gestures of body language.

14. How does noise disturb the communication process ?

15. Write two functions of Business Letters.

16. What do you mean by presentation ?

17. What do you mean by word formation ?

18. Discuss about rhetorical devices.

VUG(I)-BCA (F-1002)

Basic Math

2023-26 (Voc)

Full Marks : 70

Time : 3 hours

Answer from **both** the Sections as directed.

The figures in the right-hand margin indicate marks.

Candidates are required to give their answers in their own words as far as practicable.

SECTION–A

(Long Answer Type Questions)

Answer any *four* questions of the following: 10×4

1. If $y = (\sin^{-1} x)^2$, show that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2 y_n = 0$. Hence find $(y_n)_0$.

2. Expand $f(x, y) = \tan^{-1}\left(\frac{y}{x}\right)$ in powers of $(x-1)$ and $(y-1)$ upto third-degree term. Hence compute $f(1{\cdot}1,\ 0{\cdot}9)$ approximately.

(*Turn Over*)

3. Discuss the maxima and minima of $f(x,y)=x^3y^2(1-x-y)$.

4. Find the surface of the solid formed by revolving the cardiod $r=a(1+\cos\theta)$ about the initial line.

5. Find the area of the region in the first quadrant enclosed by the x-axis, the line $y=x$ and the circle $x^2+y^2=32$.

6. Find the volume of the solid formed by the revolution about the y-axis of the part of the parabola $y^2=4ax$ cut off by the latus rectum.

7. Solve the equation:

$$\frac{d^2y}{dx^2}-2\frac{dy}{dx}+y=xe^x\cos x.$$

8. Solve the equation:

$$\frac{\partial^2 z}{\partial x^2}+\frac{\partial^2 z}{\partial x\partial y}-6\frac{\partial^2 z}{\partial y^2}=\cos(2x+y).$$

(Continued)

SECTION–B

(Short Answer Type Questions)

Answer *all* questions of the following: 3×10

9. Find the *n*th derivative of $e^{2x}\cos^2 x \sin x$.

10. If $y^{1/m} + y^{-1/m} = 2x$, prove that

$$(x^2-1)y_{n+2} + (2n+1)y_{n+1} + (n^2 - m^2)y_n = 0.$$

11. If $u = \sin^{-1}\left(\dfrac{x+2y+3z}{x^8+y^8+z^8}\right)$, find the value of

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}.$$

12. Expand log sinx in powers of $(x-2)$ by Taylor's theorem.

13. Find the maximum & minimum value for the function $y = x^3 - 3x^2 + 5$.

14. Find the percentage error in measuring the volume of a rectangular box when the error of 1% is made in measuring each side.

15. Find the length of one arch of the cycloid $x = a(t - \sin t)$, $y = a(1 - \cos t)$.

16. Find the volume of a sphere of radius a.

17. Solve $(x^2 - y^2)dx - xydy = 0$.

18. Solve $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial x \partial y} = \cos x \cos 2y$.

VUG(I)-BCA(F-1003) /
(BCS-103)-BPM

2023-26 (Voc)

Full Marks : 70

Time : 3 hours

Answer from **both** the Sections as directed.

The figures in the right-hand margin indicate marks.

Candidates are required to give their answers in their own words as far as practicable.

SECTION–A

Answer any *four*: 10 × 4

1. Define Business Environment. Bring out the difference between Micro and Macro environment.

2. Explain co-operative organization. Write its advantage.

3. Explain the term "organization structure". What is its advantage ?

4. Write in detail about Human relation theory in organization.

5. Define Management. What are the functions of management ? Explain.

6. What is planning ? Explain the different types of planning.

7. Explain Leadership. What are the different styles of leadership ?

8. Explain in detail the communication process.

SECTION–B

Answer any *all*, 03 marks each: 3 × 10

9. What is HUF ? Explain in short.

10. What is informal organization ?

11. What is MBO ? Explain the concept.

12. Define Motivation.

13. What is Decentralization ?

14. What are the different resources in management ?

15. What are different types of control ?

16. What are the three essential skills of a manager ?

17. Explain the meaning of unity of command.

18. Explain the meaning of unity of direction.

VUG(I)-BCA(F-1004)-
Info. To Comp. Sc

2023-26 (Voc)

Full Marks : 70 40

Time : 3 hours

Answer from **both** the Groups as directed.

The figures in the right-hand margin indicate marks.

Candidates are required to give their answers in their own words as far as practicable.

GROUP—A

Answer any *four* questions: 10 × 4

1. Explain about the classification of computer.

2. (*a*) Write about Logic gates.

   (*b*) Write about Boolean algebra.

3. Write and explain the Von-neuman Architecture.

4. Write notes on:

   (*a*) Computer output microfilm

   (*b*) Digital Camera

5. Write and explain the types of operating system.

6. Discuss:

   (*a*) Pseudocode

   (*b*) Algorithm

7. Write and explain Network topology.

8. (*a*) Write about Email.

   (*b*) Write about Viruses.

GROUP—B

Answer any *all* questions: 3×10

9. What is Computer ?

10. What is Number system ?

(Continued)

11. What is RAM and ROM ?

12. What is Optical disk ?

13. What is Hardware, Software and firmware ?

14. What is the difference between internet and intranet ?

15. What is Operating system ?

16. What is Transmission media ?

17. What is Customize software ?

18. Write about communication protocol.

3x3=9

2x10=20

VUG(1)-BCA(C-1005)-
Prob. For Inc

2023-26 (Voc)

Full Marks : 70

Time : 3 hours

Answer from both the Groups as directed.

The figures in the right-hand margin indicate marks.

Candidates are required to give their answers in their own words as far as practicable.

GROUP—A

Answer any *four* questions. Each question carry equal marks: 10 × 4

1. (*a*) What is *C* ? Why *C* is known as structured programming language ?

   (*b*) Explain different sections of *C*-program in brief.

2. What is data type ? Explain different data types used in *C* and how they declared.

3. What are different types of decision making statements in *C* ? Write their syntax with suitable example of each.

4. What is Array ? Write a program in *C* to access elements of one dimensional array by using pointer.

5. What is recursion ? Write a program to print Fibonacci series upto *n* number of terms by using Recursion.

6. (*a*) What is User Defined Function ? Explain Function prototype of different types of user defined functions.

   (*b*) Write a program in C to show function with argument and return value.

7. What is structure ? Write a program in *C* to show how structure can be passed into a function.

8. What is File ? Explain different operations on file with appropriate functions for the same.

(*Continued*)

GROUP—B

Answer *all* questions. Each question carry equal marks: 3×10

9. What are identifiers and literals ?

10. What is operator ? Explain Unary operators and why it is called so.

11. What is Dynamic Array ? How dynamic array can be declared in *C* ?

12. What is type casting of a variable ? Why and when it is used ?

13. Differentiate with example between Gets( ) and Scanf( ) with reference of string.

14. What is String in C ? Write the syntax and use of Dtrcat( ), Strcpy( ) and strcmp ( ) string manipulations functions.

15. Explain the Scope, Visibility and Lifetime of a variable.

16. What is Signature of a function ? Explain with suitable example.

17. What is UNION ? How UNION is different from STRUCTURE ?

18. What is pointer ? Write advantages of pointer in C.